Control of Computer Networks (XC), Degree in Informatics Engineering		8/4/2024	Spring 2024
FIRST NAME:	**SURNAMES:**	**GROUP:**	**ID:**

Duration: 1h30m. The test will be collected in 25 minutes. Please reply on this page.

Test. (3.5 points) Questions score half if there is one error and 0 if there is more than one.

1. About IP networks:
- ☐ The network 10.0.0.0/8 is class A.
- ☐ The network 100.0.0.0/16 is class B.
- ☐ The network 192.168.255.0/24 is private.
- ☐ The network 200.0.0.0/24 is private.

2. Which of the following addresses can be a subnet address?
- ☐ 200.2.65.0/20
- ☐ 200.2.65.0/24
- ☐ 200.2.65.32/27
- ☐ 200.2.65.64/27

3. About the IP protocol:
- ☐ The ICMP message goes in the data field of an IP packet over the Internet.
- ☐ The header of an IP packet has a field to detect errors that occur in the IP header.
- ☐ The header of an IP packet has a field to detect errors that occur throughout the IP packet.
- ☐ The fragmentation of an IP packet does not occur if the DF flag (“DO NOT FRAGMENT”) is active in the IP packet header.

4. About the IP protocol:
- ☐ A device with two interfaces connected to two nets can receive packets on one and forward packets to the other with "IP forwarding" active.
- ☐ The header of the IP packets is modified in each router by putting the IP address of the next router (gateway).
- ☐ IP packets exchanged between two IP addresses always follow the same path to reach the destination.
- ☐ The IP protocol allows packets to be transported between two terminal devices (hosts) but packets can be lost or arrive out of order.

5. About the ARP protocol:
- ☐ An ARP packet with TTL=2 allows two-hop IP interface discovery.
- ☐ Clients send messages (unicast) to the IP address from which we want to discover its Ethernet MAC address.
- ☐ More than one device can respond to an ARP REQ message, but it indicates duplicate IP.
- ☐ A device connected to an Ethernet network must respond to ARP requests referring to the interface’s IP address connected to the network.

6. About the DHCP protocol:
- ☐ The client sends DHCPDISCOVER and DHCPREQUEST messages to IP address 255.255.255.255.
- ☐ It serves exclusively to configure the IP address and netmask.
- ☐ At least one DHCP server must be accessible on each network.
- ☐ To keep a dynamically assigned IP address, this must be renewed before it expires.

7. About the ICMP protocol:
- ☐ ARP uses ICMP broadcast packets.
- ☐ If there is no information in a routing table to reach the destination, an error message “Destination host unreachable” is generated.
- ☐ Allows to generate error messages.
- ☐ ICMP packets do not pass through a router with PNAT (PAT).

8. About RIP version 2:
- ☐ Each router sends route updates to networks directly connected to the router over RIP-enabled interfaces.
- ☐ It distributes updates to all routers in directly connected networks or not.
- ☐ The "split horizon" method serves to reduce the "count to infinity" effect.
- ☐ RIP messages from one router reach all routers on the network.

9. In a routing table (as shown by the command: route -n):
- ☐ Entries in the "Gateway" column can be empty or have the IP address of the local interface through which the IP packet should be sent.
- ☐ The entries in the "Gateway" column can be empty or have the IP address of the next router to forward the IP packet (1st hop).
- ☐ Applies the first matching table row.
- ☐ Applies the row with the network that matches the longest number of network bits (Longest Prefix Match, LPM).

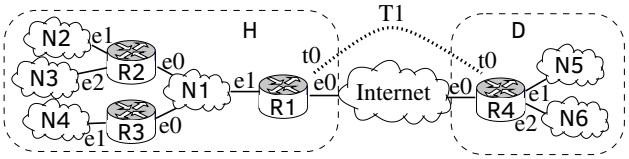
10. About NAT address translation with ports (Port Address Translation, PAT or PNAT). DNAT stands for Destination NAT:
- ☐ PAT changes the source IP address and port on packets leaving a private network to the public, while keeping destination IP address and port.
- ☐ PAT changes the source IP address and port on packets from the public network to a private, while keeping destination IP address and port.
- ☐ DNAT allows a router to redirect incoming network connections (initiated by a client on the public network), by translating the destination IP address of the packets, to allow access to servers within a private network from outside.
- ☐ DNAT translates IP addresses and ports of packet exchanges initiated by clients on a private network towards Internet servers.

Control de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	08/04/2024	Primavera 2024
Name	Surname	Group	DNI

Duration: 1h30m. The quiz will be collected in 25 minutes. Answer the problems in the same sheet.

Problem 1.

4.5 points.



N1: 200.0.0.0/29
N2: 200.0.0.144/28
N3: 200.0.0.176/28
N4: 200.0.0.192/26
T1: 192.168.0.0/24
N5: 172.16.0.0/24
N6: 172.16.1.0/24
R1-e0: 80.0.0.2
R4-e0: 90.0.0.2

In the network of the figure there is a headquarters H and a delegation D connected by an IPIP tunnel (T1). All subnets of H belong to the range 200.0.0.0/24. The network is configured so that all connections go out to the Internet through router R1 of H. For example, a connection from an N5 host will go through T1 to get out to the Internet. All routers in H and D use RIP version 2 with split-horizon and class-based route summarization. Note that *hosts* in questions 3 and 4 do not include *routers*.

1. (1 point) Complete the routing table for R1 and R4 once RIP has converged. Use as many rows as you need. In the metrics column (M) put the RIP metric. For the gateway column, assume that the routers have been assigned the numerically smallest IP. When there is more than one router in a network, IPs are assigned starting with the router with the smallest lexicographic name. For instance, R1 IP in N1 is the numerically smallest IP that can be assigned in N1.

R1

Destination/mask	Gateway	If	M
80.0.0.1/32	-	e0	1

R4

Destination/mask	Gateway	If	M
90.0.0.1/32	-	e0	1

2. (0.5 points) Say what will be the content of the update messages that will be sent by R1 on the indicated interfaces once RIP has converged. Use the notation: (destination/mask, metric).

e1	

t0	

3. (1 point) We want to expand H by adding another subnet that has the maximum number of hosts (taking IPs from 200.0.0.0/24 and not changing the existing subnets). Give the following information about the new subnet. Justify the answer.

network address/mask	broadcast address	max hosts

4. (0.5 points) We want to divide N4 into smaller networks so that there can be at least 4 hosts and 1 router in each subnet. Say the maximum number of subnets (n) could be divided into, and give the following information for the subnet with the largest numerical value.

	numerically largest subnet	
n	**network address/mask**	**max hosts**

5. (0.75 points) Hosts A and B have the largest IP that can be assigned at N6 and N3, respectively. A pings B. Say if NAT will be needed and what will be the IPs and protocol in the IP header of the echo request when it arrives at R1 via e0 and when it leaves R1 via e1.

	external header			internal header		
e0 in	**source address**	**destination address**	**protocol**	**source address**	**destination address**	**protocol**

e1 out	**source address**	**destination address**	**protocol**

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6. (0.75 points) Host A now pings IP 8.8.8.8. Say if NAT will be needed and what will be the IPs and protocol of the IP header of the echo request when it arrives at R1 via e0 and when it leaves R1 via e0.

	external header			internal header		
e0 in	**source address**	**destination address**	**protocol**	**source address**	**destination address**	**protocol**

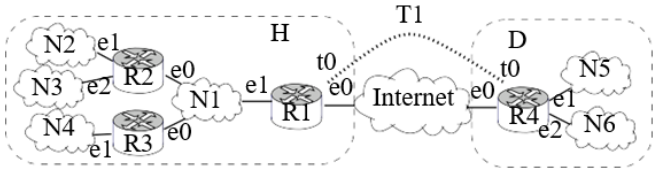
e0 out	**source address**	**destination address**	**protocol**

Midterm. Xarxes de Computadors (XC). Grau en Enginyeria Informàtica		08/04/2024	Spring 2024
NAME (UPPER-CASE):	LAST NAME / SURNAME (UPPER-CASE):	GROUP:	DNI/NIE:

Answer on the same sheet.

Problem 2 (2 points)

The figure shows the network configuration and the allocated IP addresses. Network H uses public addresses from prefix H and network D uses private addresses from prefix D. All connections to Internet go through R1; that is, networks N5 and N6 communicate with H and the Internet via the tunnel T1.



H: 200.0.0.0/24
N1: 200.0.0.0/29
N2: 200.0.0.144/28
N3: 200.0.0.176/28
N4: 200.0.0.192/26
T1: 192.168.0.0/24

N5: 172.16.0.0/24
N6: 172.16.1.0/24
D: 172.16.0.0/16
R1−e0: 80.0.0.2
R4−e0: 90.0.0.2

Access Control List (ACL) are defined in some router interfaces. Router interfaces without ACL allow all traffic in both directions.

rule	Interface	IN/OUT	Source IP	Source port	Destination IP	Dest. port	Protocol	Action
1	R3e0	IN	ANY	>=1024	N4	<1024	TCP	Accept
1	R3e0	OUT	N4	<1024	ANY	>=1024	TCP	Accept
2	R3e0	IN	ANY	53 (dns)	200.0.0.196/32	>=1024	TCP/UDP	Accept
2	R3e0	OUT	200.0.0.196/32	>=1024	ANY	53 (dns)	TCP/UDP	Accept
3	R3e0	IN/OUT	ANY		ANY		ICMP	Accept
	R3e0	IN/OUT	ANY	ANY	ANY	ANY	ANY	Deny
4	R1e1	IN	200.0.0.5/32	22(ssh)	ANY	>=1024	TCP	Accept
4	R1e1	OUT	ANY	>=1024	200.0.0.5/32	22 (ssh)	TCP	Accept
5	R1e1	IN	ANY	22 (ssh)	ANY	>=1024	TCP	Deny
5	R1e1	OUT	ANY	>=1024	ANY	22 (ssh)	TCP	Deny
6	R1e1	IN/OUT	ANY	ANY	ANY	ANY	TCP/UDP/ICMP	Accept
	R1e1	IN/OUT	ANY	ANY	ANY	ANY	ANY	Deny

a) (1 point) Answer the following questions and write TRUE or FALSE to each of the sentences below.
Any TCP client in H, D and Internet may access the servers in N4.
TCP clients in N4 may communicate with any server in the Internet.
TCP clients in N4 may access servers in D.
The host 200.0.0.196 is the only client in N4 with access to the Internet.
Hosts in N6 may establish an SSH (TCP:22) connection with server 200.0.0.5.
Rules 4, 5 and 6 forbid SSH connections between N2 and N4.
With rules 4 and 5 a unique SSH server is allowed in N1.
Rule 6 allows all TCP, UDP and ICMP traffic but forbids any other protocol.
It is feasible to establish an IPinIP tunnel between R2 and an external router in the Internet.
It is feasible to establish a VPN from a host in N3 with an external VPN server in the Internet.

b) (1 point) Configure the Access Control List (ACL) rule at the corresponding interface so that: a) host in N5 are TCP clients of servers located in the Internet and in H, and simultaneously may be TCP servers of clients in N6 only; b) host in N5 only answer to "*ping*" from hosts in N6.

rule	Interface	IN/OUT	Source IP	Source port	Destination IP	Dest. port	Protocol	Action
a								
a								